

NVIDIA Jetson Orin Nano 8GB

Local Multimodal Personal Assistant

Installation guide: boot to NVMe, local + cloud models, voice, Home Assistant, and the OpenClaw agent harness - secured, outbound-only, with human-in-the-loop approval.

Key decisions captured in this build

- Install: JetPack 7.2 'Jetson ISO' on a USB stick installs Jetson Linux directly onto the NVMe (no SD clone, no x86 Linux host).
- Local runtime: llama.cpp llama-server (OpenAI-compatible), 3-4B Q4 multimodal model.
- Optional cloud offload: GitHub Copilot provider, latest Claude Opus as preferred model.
- Agent harness: OpenClaw (sandboxed), WhatsApp Cloud API channel, admin via WireGuard.
- Security: outbound-only (no inbound ports), DM pairing, no payment credentials, explicit approval for high-impact actions.

Version 1.0 - generated 2026-06-22. Pin versions in versions.env; treat this as living documentation.

1. Overview & Architecture

This guide builds an always-on home assistant on a Jetson Orin Nano 8GB with a 500GB NVMe SSD. It serves a frequently-updated multimodal model behind an OpenAI-compatible endpoint and layers the OpenClaw agent harness on top, with strong security defaults.

Components

- llama.cpp (llama-server): local OpenAI-compatible inference for a 3-4B quantized vision model.
- GitHub Copilot provider (optional): cloud models (Claude Opus) to offload the Jetson for heavy reasoning.
- Open WebUI: browser chat interface.
- Wyoming Whisper (STT) + Piper (TTS): voice, integrated with Home Assistant Assist.
- OpenClaw: agentic harness (skills, browser, messaging) running sandboxed.
- WireGuard + VPS: outbound-only admin access and WhatsApp webhook relay.

Architecture (data flow)

```
WhatsApp Cloud API --> VPS (webhook + WireGuard server)
                        | (relayed over WireGuard tunnel)
                        v
Jetson Orin Nano 8GB (outbound-only; no inbound router ports)
- llama-server :8080 -> OpenAI /v1 (local model)
- Open WebUI   :3000
- Whisper :10300 Piper :10200 -> Home Assistant Assist
- OpenClaw Gateway (sandboxed)
  primary = github-copilot/claude-opus-4.8 (cloud)
  fallback = llamacpp/<local model>         (offline)
```

Design principles

- Outbound-only: the device initiates all connections; no inbound ports on the home network.
- Approval gates: high-impact actions (orders, payments) require explicit human confirmation.
- Least privilege: the agent holds no payment/order credentials; sandboxed tool execution.
- Version pinning: reproducible deployment via versions.env; avoid 'latest' in production.

2. Hardware Checklist & Memory Budget

Hardware

- Jetson Orin Nano 8GB Developer Kit (Super-capable; JetPack 7.2 / L4T r39.2).
- NVMe M.2 SSD 500GB (PCIe Gen3) installed in the M.2 Key-M slot.
- USB stick (≥ 16 GB) for the one-time JetPack installer (throwaway boot media).
- DisplayPort monitor + USB keyboard for the one-time install (no video over USB-C); headless via SSH afterwards.
- Active cooling (fan) - required for sustained inference.
- Included 19V power supply (barrel jack); MAXN SUPER needs the full supply - don't rely on weak USB-C PD.
- Ethernet (recommended) for a stable headless server.

Memory budget (8GB unified - plan carefully)

Component	Approx. RAM
OS (headless)	~1.5 GB
Local model (3-4B Q4)	3-4 GB
Whisper (base)	0.5-1 GB
Piper (TTS)	<0.2 GB
Open WebUI	~0.3 GB
OpenClaw (Node)	0.3-0.5 GB
Chromium (browser, on-demand)	0.5-1.5 GB

[!] 8GB is the hard constraint

Everything active at once exceeds 8GB. Use a 3-4B local model (not 7B), run Chromium on-demand only, keep 8-16GB swap on NVMe, and prefer the cloud (Copilot) model for heavy work. Serialize peaks (voice OR browser-agent, not both).

3. Phase 1 - Install JetPack to the NVMe (USB-ISO)

Since JetPack 7.2 (L4T r39.2, mid-2026), NVIDIA ships a 'Jetson ISO' installer that you write to a USB stick. You boot the Jetson once from that USB stick and the installer updates the QSPI firmware and installs Jetson Linux DIRECTLY onto the NVMe SSD - no SD card, no rootfs clone, no x86 Linux host. The USB stick is throwaway boot media (like a DVD); remove it afterwards.

On Windows: write the ISO to a USB stick

1) Download the Jetson ISO from NVIDIA (developer.nvidia.com -> JetPack downloads -> Jetson ISO r39.2). 2) Write it to a >=16GB USB stick with balenaEtcher (Flash from file -> Select target = the USB stick -> Flash). Verify the download is complete (size / SHA256). The firmware update is baked into the ISO - no internet needed for it.

[!] No video over USB-C

The Orin Nano Developer Kit outputs display only over its DisplayPort connector (no HDMI port; USB-C does NOT carry video). The UEFI firmware prompt cannot be driven over USB-C, so you need a DisplayPort monitor + USB keyboard (or a USB-to-TTL serial cable) for the one-time install. Afterwards the device runs fully headless via SSH.

On the Jetson: boot the USB and install

- Attach a DisplayPort monitor + USB keyboard, Ethernet, the NVMe, and the USB stick, then connect the included 19V power supply.
- Press Esc at the NVIDIA splash -> Boot Manager -> select the USB stick.
- Press Y within 30 seconds to accept the QSPI firmware capsule update (the most-missed step; the install fails later if skipped). It runs in two passes with reboots.
- At the GRUB menu choose 'Install Jetson ISO r39.2', select the NVMe SSD as the target (not the USB, not a microSD), and confirm (this erases the NVMe).
- Reboot when prompted and remove the USB stick - the system now runs entirely from the NVMe.

On first boot from the NVMe, complete the Ubuntu setup (EULA, language, network, create your user + hostname), then set the power mode to MAXN SUPER. SSH (openssh-server) is enabled by default on L4T, so the device is reachable over the network immediately after this step.

Note

If the firmware is older than 36.x, the installer's capsule update brings it current first. If your dev kit shipped with factory firmware too old for JetPack 7, run NVIDIA's 'JetPack 6.x update path' once, then retry the JP7 installer.

4. Phase 2 - System Configuration

Run on the device after booting from NVMe. See scripts/01-system-config.sh and scripts/02-swap.sh in the repo.

Power, cooling, headless

```
sudo nvpmode1 -m 0          # MAXN SUPER
sudo jetson_clocks          # max clocks
sudo systemctl set-default multi-user.target # disable desktop GUI (frees RAM)
```

Base packages + Node 24 (for OpenClaw)

```
sudo apt install -y build-essential git cmake curl ca-certificates
curl -fsSL https://deb.nodesource.com/setup_24.x | sudo -E bash -
sudo apt install -y nodejs
```

Swap on NVMe (OOM safety net)

```
./scripts/02-swap.sh 12      # creates /swapfile on the NVMe, swappiness=10
```

- Static IP or DHCP reservation for a stable headless server.
- SSH is enabled by default on JetPack; manage everything remotely.

5. Phase 3 - Docker & jetson-containers

JetPack ships Docker with the NVIDIA container runtime. Prefer prebuilt Jetson images (dustynv / jetson-containers) over building CUDA from source.

```
docker info | grep -i runtime          # confirm 'nvidia' runtime
git clone https://github.com/dusty-nv/jetson-containers
bash jetson-containers/install.sh
```

Tip

After the JetPack install the NVMe is the root filesystem (/) - the only drive - so model weights (/opt/models) and Docker data (/var/lib/docker) already live on it. Just don't fill it up. Pin image tags in versions.env.

6. Phase 4 - Local Inference (llama.cpp)

llama-server provides an OpenAI-compatible /v1 API for the local multimodal model (privacy/offline fallback). For vision you need the model GGUF and its matching --mmproj projector.

Model choice (balance)

- Gemma 3 4B (Q4_K_M) + mmproj, or Qwen2.5-VL 3B (Q4) + mmproj.
- Download GGUF from a trusted source; verify checksums; pin the filename.
- Store under /opt/models (on the NVMe root; no separate /mnt/nvme exists).

Run (see compose/docker-compose.yml)

```
llama-server -m /models/gemma-3-4b-it-Q4_K_M.gguf \  
  --mmproj /models/gemma-3-4b-mmproj.gguf \  
  --host 0.0.0.0 --port 8080 -ngl 99 --ctx-size 4096  
  
curl http://127.0.0.1:8080/v1/models # health check
```

7. Phase 5 - Open WebUI

A browser chat UI pointed at the local endpoint. Brought up by docker-compose.

```
# environment (compose):  
OPENAI_API_BASE_URL=http://llamacpp:8080/v1  
OPENAI_API_KEY=sk-no-key-required  
# UI on http://<jetson-ip>:3000
```

Security

Bind UI/endpoint ports to localhost or the WireGuard interface in production - do not expose them to the LAN/internet directly.

8. Phase 6 - Voice (Whisper + Piper)

Wyoming services provide speech-to-text and text-to-speech that Home Assistant Assist can discover.

```
wyoming-whisper  --model base  --language nl    # :10300
wyoming-piper    --voice nl_NL-mls-medium      # :10200
```

- Load Whisper base/tiny and on-demand to conserve the 8GB budget.
- Both run as containers in docker-compose.yml.

9. Phase 7 - Home Assistant Assist

- Add Wyoming integrations in HA pointing at <jetson-ip>:10300 (Whisper) and :10200 (Piper).
- Create an Assist pipeline: Whisper (STT) -> Conversation agent (LLM) -> Piper (TTS).
- For the LLM, use the Extended OpenAI Conversation custom integration (HACS) so you can set a custom base_url to http://<jetson-ip>:8080/v1.

Approval boundary

HA voice must not trigger high-impact agent actions (orders/payments) without the OpenClaw approval gate.
Keep HA on the LAN/WireGuard network only.

10. Phase 8 - Connectivity (Outbound-only)

No inbound ports on the home network. The Jetson dials out; interaction returns over the established connections.

Admin: WireGuard

- Run a WG client on the Jetson that dials a WG server on your VPS.
- You reach the Jetson over the private tunnel; the home router opens no ports.
- Keep PersistentKeepalive=25 on the Jetson side. See wireguard/ in the repo.

Agent channel: WhatsApp Cloud API

- The official Cloud API delivers inbound messages via a webhook (inbound by nature).
- Host the webhook receiver on the VPS, verify the Meta signature, and relay to the Jetson over WireGuard.
- Result: the Jetson / home network opens no inbound ports.

11. Phase 9 - OpenClaw Agent Harness

OpenClaw (github.com/openclaw/openclaw, MIT) is a self-hosted personal assistant that bridges chat apps to LLMs with skills, browser control and shell access. Microsoft's 'Scout' is built on OpenClaw. Runtime: Node 24.

Install (pin a version; do not blind-curl)

```
npm install -g openclaw@latest
openclaw onboard --install-daemon      # installs the Gateway systemd service
openclaw gateway status
```

Security configuration (NOT the defaults)

[!] Default = full host access for the main session

You must override this. Sandbox non-main sessions (Docker), keep sandbox egress off (network: none), require pairing for unknown senders, and never enable tools.elevated (it bypasses the sandbox).

```
# config/openclaw/config.json5 (excerpt)
agents.defaults.sandbox = {
  mode: 'non-main', backend: 'docker', scope: 'session',
  docker: { network: 'none' },      # no egress for sandboxed shell/code
}
tools = { profile: 'coding', alsoAllow: ['browser'] } # capable + browser on
browser = { enabled: true }           # isolated 'openclaw' profile (see below)
logging = { redactSensitive: 'tools' }
channels.whatsapp.dmPolicy = 'pairing'
openclaw pairing approve whatsapp <code> # approve a sender
```

The browser is enabled - but isolated

- The agent uses a separate 'openclaw' browser profile, NOT your personal browser.
- That profile has no logged-in webshops and no saved payment methods, so the agent can browse and verify freely but cannot complete a purchase with your cards.
- Do not switch the agent to the 'user' profile (your real signed-in Chrome).
- Give OpenClaw NO payment/order credentials (it can draft an order; you execute it).
- Run 'openclaw security audit' (add --deep / --fix) after any config or exposure change.
- Read the official 'Gateway exposure runbook' before any remote exposure.
- ARM64: Node + Playwright/Chromium work; watch for x86-assuming skills.

12. Phase 10 - Optional: GitHub Copilot Cloud Offload

Offload heavy reasoning to cloud models via your GitHub Copilot subscription, keeping the local model as an offline/private fallback. This relaxes the 8GB pressure. OpenClaw supports a github-copilot/* provider natively.

Install the Copilot SDK harness plugin

```
openclaw plugins install @openclaw/copilot # ~260MB incl. CLI binary
# verify the linux-arm64 Copilot CLI binary loads on the Jetson:
openclaw doctor
```

Preferred model = latest Claude Opus

```
// config/openclaw/config.json5 (excerpt)
agents.defaults.model = {
  primary: 'github-copilot/claude-opus-4.8', // confirm id: openclaw models list
  fallbacks: ['llamacpp/gemma-3-4b']
}
models: {
  'github-copilot/claude-opus-4.8': { agentRuntime: { id: 'copilot' } },
  aliases: { 'mijn-default': 'github-copilot/claude-opus-4.8' }
}
```

Choosing your model (three levels)

- Config: agents.defaults.model.primary + fallbacks (persistent default).
- CLI: openclaw models list / set, plus aliases.
- In chat: /model picker (per-session, pinned).

13. Security Model & Approval Gates

Principle: maximise useful autonomy, minimise blast radius. The assistant should resolve the routine 95% on its own and stop only for the consequential 5%. It must do a genuine check-in for sensitive actions - and never try to force, retry around, or social-engineer its way past a denied/pending approval.

Autonomy tiers

- Tier A - Autonomous (no prompt): search, browse (isolated profile), summarise, draft replies, run code in the sandbox, install Python/npm packages in the sandbox, read Home Assistant sensors, manage its own workspace files.
- Tier B - Act + notify: reversible local changes and non-critical home control (e.g. lights); the agent acts and tells you what it did.
- Tier C - Approval required (hard stop): spending money, orders/purchases/payments, messaging or emailing third parties on your behalf, posting publicly, deleting data irreversibly, changing credentials/security, controlling critical devices, or running outside the sandbox.

Why a denied approval actually holds

- Least privilege: the agent holds NO payment/order credentials, so a Tier C action cannot physically complete without you - the gate is structural, not just a prompt instruction.
- Isolated browser profile: no saved cards or logged-in webshops, so 'just check out on the site' is not possible.
- Sandbox with no egress (network: none) for shell/code; tools.elevated stays off.
- The agent is instructed to surface the blocker and wait, not to brute-force or seek a workaround.

Controls in this build

- Outbound-only networking; no inbound router ports (WireGuard + VPS relay).
- OpenClaw sandboxed (Docker), sandbox egress network: none, DM pairing on.
- Browser enabled but isolated ('openclaw' profile) - capable yet credential-free.
- No payment/order credentials available to the agent (it drafts; you execute).
- Secrets in .env (gitignored); WireGuard private keys generated on-device.
- Audit: run 'openclaw security audit' (--deep/--fix) and 'openclaw doctor'; keep redacted logs (logging.redactSensitive: 'tools').

Golden rule

Agents must never place orders or take consequential actions without explicit prior approval - and must never try to force or bypass that approval. Enforced by least privilege (no credentials) + sandbox + isolated browser, not by prompt instructions alone.

Borrowed from Microsoft Scout & recognised frameworks

- Scout identity/SSO -> personal analog: strict DM pairing + allowlists (only you can drive the bot); one trusted operator per gateway.
- Scout policy/governance -> tool allowlists + autonomy tiers + 'openclaw security audit --fix' (flips open policies to allowlists).
- Scout audit/observability -> redacted tool logs + periodic audit/doctor runs.
- OWASP LLM/Agentic Top 10 -> mitigate excessive agency (least privilege + approval) and prompt injection (treat inbound DMs AND web page content as untrusted).

- NIST AI RMF + least-privilege/egress control -> sandbox network: none, no inbound ports, deliberate version pinning.

14. Verification & Benchmarking

```
lsblk                                # root on nvme0n1p1
nvpmodel -q                         # MAXN SUPER
curl http://127.0.0.1:8080/v1/models
docker compose ps                   # all services up
openclaw gateway status
# quick local inference latency:
time curl -s http://127.0.0.1:8080/v1/chat/completions \
  -H 'Content-Type: application/json' \
  -d '{"model":"local","messages":[{"role":"user","content":"hi"}]}'
```


15. Maintenance & Updates

- Update models: download a new GGUF, update the filename in compose + versions.env, restart llamacpp.
- Update OpenClaw: `npm update -g openclaw` (then 'openclaw doctor'); pin the version in versions.env.
- Update containers: bump tags in versions.env deliberately, then `docker compose pull && up -d`.
- Keep the OS current: `sudo apt update && sudo apt full-upgrade` (includes firmware).
- Commit every change to the deployment repo so the running state is captured.

16. Troubleshooting

Symptom	Fix
NVMe won't boot after install	Re-run the USB installer; ensure you pressed Y for the firmware capsule; check UE
OOM / killed processes	Use 3-4B model, add swap, run Chromium on-demand
pip install torch broken	Use Jetson-specific wheels / jetson-containers, not default pip
Copilot runtime missing	openclaw plugins install @openclaw/copilot; openclaw doctor
WhatsApp not receiving	Check VPS webhook + Meta signature + WireGuard tunnel
Solcast/secrets empty	Secrets in .env, never committed; check env_file mapping

Appendix - Version Pin Table

Mirror of versions.env. Update deliberately; verify on-device before bumping.

Item	Pinned value
JetPack	7.2
L4T	r39.2
llama.cpp image	dustynv/llama_cpp:r39.2.0 (verify tag)
Open WebUI	ghcr.io/open-webui/open-webui:main
Wyoming Whisper	rhasspy/wyoming-whisper
Wyoming Piper	rhasspy/wyoming-piper
Node	24
OpenClaw	pin once chosen
Cloud model	github-copilot/claude-opus-4.8
Local model	llamacpp/gemma-3-4b (Q4)